DT 111 · TOPIC 4

Statistical Methods

Probability, inference, hypothesis testing and simple regression

Mkwawa University College of Education

BSc Data Science and Artificial Intelligence

Learning outcomes

- Explain population, sample, parameter and statistic.
- Use basic probability and conditional probability concepts.
- Describe common probability distributions intuitively.
- Interpret null/alternative hypotheses, p-values and confidence intervals.
- Interpret slope, prediction and residuals in simple regression.

Why statistical methods matter

Data from samples contain variation and uncertainty. Statistical methods help us reason beyond the observed sample without pretending to know more than the evidence supports.

Population and sample

Population

- Full group of interest
- Example: all first-year students at MUCE
- Characteristics are described by parameters

Sample

- Observed subset of the population
- Used to learn about the population
- Characteristics are described by statistics

Parameter and statistic

Parameter

A numerical characteristic of a population, usually unknown.

Statistic

A numerical summary calculated from a sample.

Inference

Using sample evidence to estimate or test claims about the population.

Example: probability and complements

A class has 40 students and 10 are absent.

Absent

10 / 40

P(absent) = 0.25

Present

30 / 40

P(present) = 0.75

Complement

1 - 0.25

P(present) = 1 - P(absent)

Core probability ideas

Outcome

One possible result.

Event

An outcome or set of outcomes of interest.

Probability

A number from 0 to 1 describing how likely an event is.

Random variable

A numerical representation of uncertain outcomes.

Discrete and continuous random variables

Discrete

- Takes countable values
- Example: number of absent students
- Often represented by counts

Continuous

- Can take values on a continuum
- Example: temperature or weight
- Often represented by measurements

Three useful distribution ideas

Bernoulli

One trial with two outcomes, such as pass/fail.

Binomial

Number of successes in a fixed number of similar Bernoulli trials.

Normal

Continuous bell-shaped model used in many measurement contexts.

Hypothesis testing

A structured way to ask whether observed sample evidence is unusually inconsistent with a default population claim.

Example: hypothesis testing

1

Question

Did a new teaching method change average score?

2

H0

No population mean difference

3

H1

A population mean difference exists

4

Sample

Observed means differ

5

Test

Is that difference unusually large under H0?

Null and alternative hypotheses

Null hypothesis (H0)

- Default claim
- Often “no difference” or “no effect”
- Provides a reference model for the test

Alternative hypothesis (H1)

- Competing claim
- Often “a difference/effect exists”
- Supported when evidence against H0 is sufficiently strong

What a p-value expresses

Definition

How unusual the observed result, or something more extreme, would be if the null hypothesis were true.

Small p-value

The observed data are less compatible with the null model.

Important limit

A p-value is not the probability that H0 is true and does not measure practical importance.

Statistical significance and practical importance

Statistical significance

- Evidence against a null model
- Influenced by sample size and variability
- Does not describe effect size by itself

Practical importance

- Is the difference large enough to matter?
- Depends on context and consequences
- Should be considered with effect size and uncertainty

Confidence intervals add uncertainty information

An interval communicates both the estimated effect and its uncertainty.

Estimated improvement

5 marks

Point estimate

95% confidence interval

2 to 8

Plausible range under the method assumptions

Simple linear regression

Regression describes how an outcome changes with a predictor and can be used for numerical prediction.

A simple regression equation

Equation

Score = 35 + 4 x Hours

A linear model

Slope

+4

Expected score change for one extra study hour

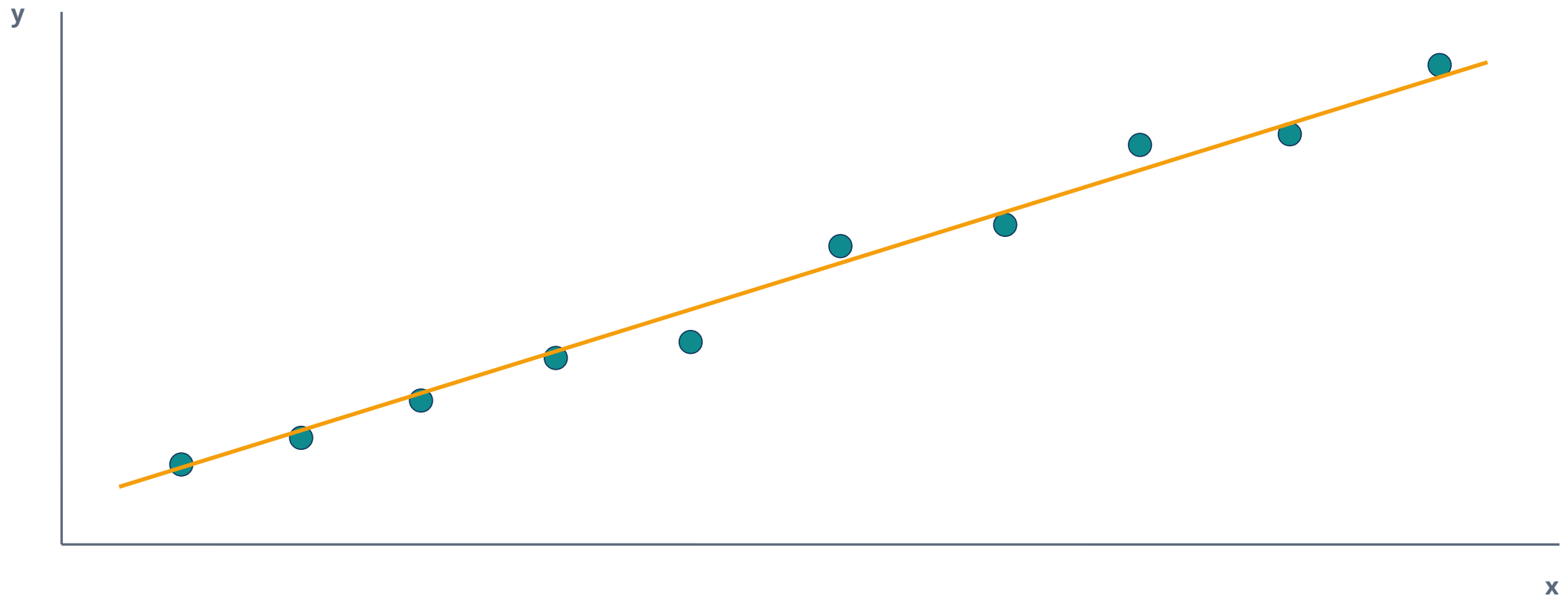
5 study hours

55

Predicted score = 35 + 4(5)

Regression line and observed data

The fitted line summarises the average linear relationship.



Residuals and model fit

Residual

Observed value - predicted value. It shows the prediction error for one observation.

R2

Describes how much variation in the outcome is explained by the fitted linear relationship.

Caution

Good fit does not prove causation or guarantee reliable predictions outside the observed range.

Association and causation

Regression can show

- Direction of a relationship
- Estimated change in outcome
- Predictions under model assumptions

Regression cannot prove alone

- That the predictor causes the outcome
- That unmeasured variables are irrelevant
- That the relationship holds in every population

Topic 4 key ideas

Statistics helps reason from samples while acknowledging uncertainty.

Probability models uncertain events; distributions describe patterns of possible values.

Hypothesis tests compare evidence with a default claim; p-values are not probabilities of hypotheses.

Confidence intervals communicate effect size and uncertainty.

Regression describes association and prediction, not automatic causation.